

1. Question from Q3-A-S2:

Given the following -

Process	Base Register	Limit Register	Logical Address from the program	Corresponding physical address	Valid?
P1	10000	1000	980		
P2	20000	2048	512		
P3	25600	1024	1100		

- For each process, calculate and fill up the column "Corresponding physical address".
- Fill-up "Valid?" column also by determine whether the memory access is valid or invalid for each process using the condition: Valid if Logical Address < Limit Register

2. Question from Q3-B-S1:

Parameter	Value
Virtual address space	16 bits
Page size	2 KB
Physical memory size	16 KB

- Calculate the number of page frames in physical memory.
- If a process generates a virtual address = 5500, find the corresponding physical address, assuming: Page 0 → Frame 2, Page 1 → Frame 7, Page 2 → Frame 1, Page 4 → Frame 4.
- What happens if the process tries to access an address that belongs to page 3?

3. Segmentation: Given Segment Table:

Segment #	Base Address	Length (Limit)
0	219	600
1	2300	14
2	90	100
3	1327	580
4	1952	96

Basic Translation: convert virtual address (2, 120) to physical address

4. Segmentation with paging:

Seg #	PT Base	Length (pages)
0	100	4
1	1000	4
2	2000	2

Page Tables:

- Seg 0: P0→F5, P1→F8, P2→F2, P3→F7
- Seg 1: P0→F10, P1→F3, P2→F9, P3→ F1
- Seg 2: P0→F6, P1→F4

Assume page size = 512 bytes

Convert virtual address (1, 3, 200) to physical address